BELL METAL INDUSTRY OF SARTHEBARI, ASSAM

History • Products • Artisans and Job Roles • Production Process • Advantages and Disadvantages

Major groups of Sarthebari bell-metal products

Utensils	Utility & ceremonial	Musical & religious
Kahi / Kanhi	Bota	Tal / cymbals
Bati	Sorai / Xorai	Bhortal
Ban-bati	Tema	Khuti-tal / Juri-tal
Ghoti / Koloh	Ban-kahi	Idols

Decorative products such as vases, lamp stands, wall clocks and showpieces are also made according to demand.

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1. Introduction to Bell Metal and Sarthebari

Bell metal is a high-tin bronze, essentially a copper–tin alloy. Technical studies commonly describe bell metal used in India as containing roughly 75–80% copper and 20–25% tin by weight. The relatively high tin content gives the alloy its characteristic hardness and strong reverberating or ringing quality, which explains its traditional use for bells, cymbals, utensils and ritual objects.

Sarthebari is one of Assam's best-known traditional metal-craft centres. The craft is organized as an artisan-based cottage industry and is closely connected with Assamese domestic life, Vaishnavite religious practice, ceremonial exchange and folk music. Government of Assam descriptions identify Sarthebari as the principal centre of the bell-metal craft and note that the technology has remained largely traditional and hand-tool based.

Important distinction
Bell metal is not the same as ordinary brass. Brass is primarily a copper–zinc alloy, whereas bell metal is a copper–tin alloy (high-tin bronze). This difference strongly affects colour, hardness, forming behaviour and sound.

2. Historical Background and Development

2.1 Antiquity of the craft

The Sarthebari tradition is regarded as very old. Government of Assam historical notes connect the craft with the seventh century A.D. and refer to the tradition that Kumar Bhaskar Varman of ancient Kamrupa presented bell-metal articles to Harshavardhana of Kanauj. The episode is often cited to show that the craft had already attained social and royal importance by that period.

- The existence of such references indicates that bell-metal ware was not merely a household commodity; it was also a prestige product suitable for royal gifts and ceremonial use.
- The craft subsequently received patronage from rulers, landed elites and religious institutions, helping it remain embedded in Assamese society.
- Another historical tradition records royal recognition of highly skilled Sarthebari artisans; official Assam material mentions Jiudhan Kahar receiving honour and revenue-free land for exceptional craftsmanship.

2.2 Development under traditional patronage

For centuries, production was organized through hereditary skill and small household or neighbourhood workshops. Knowledge of alloy handling, heating, hammering, shaping and finishing passed from experienced masters to younger assistants through practice rather than formal technical education. The survival of the craft depended on this apprenticeship-type system.

2.3 The Mahajan system before Independence

Before Independence, artisans often depended on traders or “Mahajans” from outside the craft community for scrap bell metal, working capital and consumables. The artisan mainly supplied labour. The Mahajan controlled raw-material supply and purchase conditions, and payment was made as labour charges after allowing for metal-weight loss. This arrangement made artisans economically dependent even though the production skill remained with them.

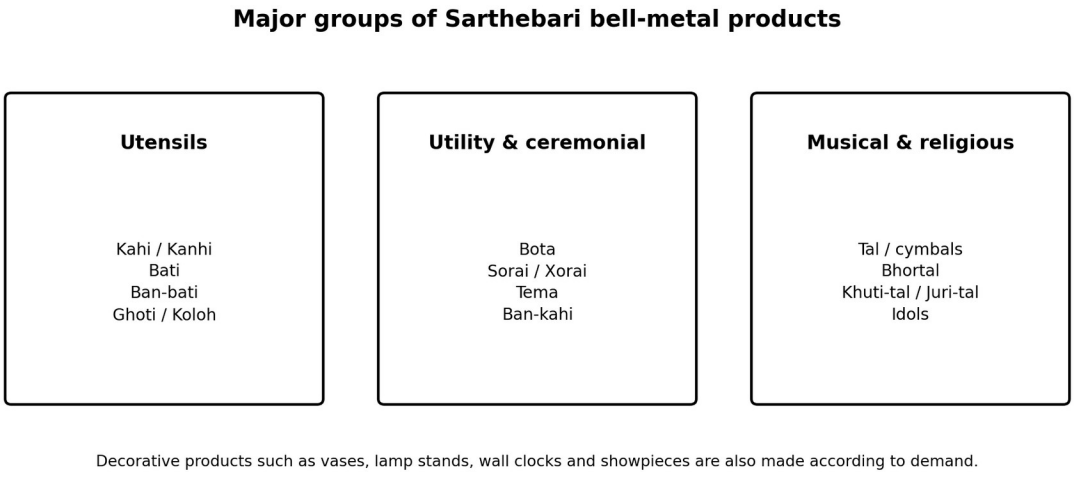
2.4 Cooperative organization from 1930

A major institutional change came in 1930 with the formation of the “Asom Samabai Kohar Sangha”, registered under the Assam Cooperative Societies Act, 1912. According to Government of Assam records, the cooperative began supplying raw material and advance money for consumables, and it fixed labour rates for different articles. This reduced the earlier domination of Mahajans and gave artisans a more organized basis for production and marketing.

2.5 Attempts at modernization

In 1957–58, the Industries Department, Government of Assam, established a five-ton-capacity rolling mill at Sarthebari. The mill could not be used effectively because of shortage of suitable skilled manpower and high operating cost. The episode illustrates a recurring issue in traditional craft modernization: introducing machinery is not sufficient unless the technology is economically viable and compatible with local skills and production scale.

3. Bell Metal Products of Sarthebari

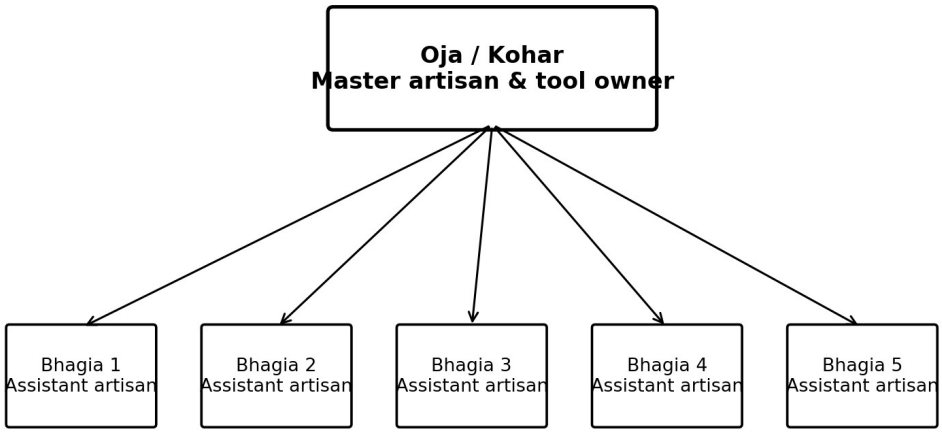


The product range combines everyday utility, ritual importance and musical use. The same object may also carry symbolic meaning; for example, the sorai/xorai is both a functional tray and an important object of Assamese welcome, respect and ceremonial presentation.

Category	Product	Typical use / significance
Utensils	Kahi/Kanhi	Dish or plate used for serving food
Utensils	Bati	Bowl for food and household use
Utensils	Ban-bati	Bowl provided with a stand or pedestal
Utensils	Ghoti / Koloh	Water vessel / container
Utility / ceremonial	Bota	Tray or stand used for offering prasad, betel nut and other items
Utility / ceremonial	Sorai / Xorai	Ceremonial tray used in religious functions and for presenting gifts/honours
Utility / ceremonial	Tema / Temi	Small container, traditionally for ornaments, betel nut or related items
Musical / religious	Tal / cymbals	Percussion instrument used in namghar worship, devotional music and folk traditions
Musical / religious	Bhortal / larger cymbal types	Large cymbals producing a strong resonant sound
Decorative / modern	Vases, lamp stands, clocks, wall pieces, idols	Produced according to changing demand and market preference

4. Artisans and Their Job Roles

The traditional production unit is called a **Sal**. It is both a workplace and a small economic partnership. The skilled master artisan is generally known as the **Kohar or Oja**, while the assisting craftsmen are known as **Bhagias**. The organization depends on cooperation because many operations—heating, holding, rotating and hammering a workpiece—must be coordinated in rhythm.



Workplace = Sal

Traditional earnings are shared; official Assam descriptions note the Oja receives a larger share because the tools belong to him.

4.1 Oja / Kohar (master artisan)

- Owns or controls the principal tools and implements of the Sal.
- Possesses the highest level of practical knowledge regarding heating temperature, hammering sequence, shape control and finishing.
- Plans the article to be produced and determines the sequence of operations.
- Directs the assistants during coordinated hammering and forming.
- Checks dimensions, symmetry, sound and surface quality.
- Often handles customer requirements, raw-material decisions and the distribution of earnings within the Sal.

4.2 Bhagias (assistant artisans)

- Assist the Oja in heating and reheating the metal.
- Hold or rotate the workpiece with tongs during hammering.
- Carry out repeated hammering under the Oja's direction.
- Perform filing, edge trimming, polishing and other finishing operations as their skill increases.
- Learn the craft through observation, repetition and participation; with experience, a Bhagia may become capable of independent work.

Traditional sharing system
Government of Assam descriptions note that the earnings of a Sal are divided among the Oja and Bhagias, with the Oja receiving about one-and-a-half times the share of a Bhagia because the Oja owns the implements and bears greater responsibility.

4.3 Indirect job roles linked to the industry

- Suppliers and collectors of broken/scrap bell metal.
- Charcoal suppliers and transporters.
- Operators and workers associated with rolling facilities that convert cast lumps into sheets or circular blanks.
- Traders, wholesalers, retailers and marketing agents.
- Persons involved in packaging, transport and sale across Assam, the North-East and neighbouring markets.

5. Raw Materials, Tools and Workplace

5.1 Main raw materials

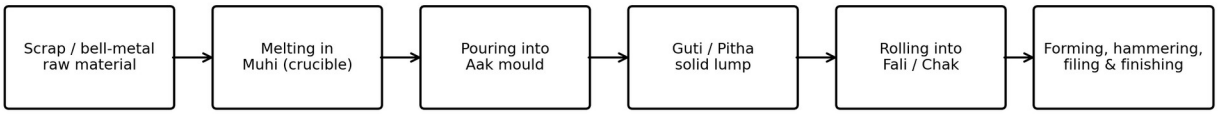
- Bell-metal scrap / recycled high-tin bronze: a major input in the traditional cluster.
- Copper and tin, directly or indirectly embodied in the recycled alloy; technically, bell metal is approximately 75–80% Cu and 20–25% Sn by weight in commonly reported compositions.
- Charcoal: the traditional fuel used extensively for heating and melting.
- Moulding material for the Aak and other locally prepared supporting materials used during casting and shaping.

5.2 Important traditional tools and equipment

Tool / local name	Function
Muhi	Crucible used for melting bell-metal scrap/alloy.
Aak	Mould into which molten metal is poured to obtain a thick circular lump/blank.
Niyari	Anvil or supporting iron surface used during hammering.
Sara	Tong used for holding hot workpieces.

Hammer(s)	Principal forming tools; different sizes are used for heavy shaping and fine finishing.
Khonta	File used to remove rough edges and improve finish.
Kun	Hand-operated lathe / rotating tool used in finishing certain articles.
Chalti and other local hand tools	Used in handling, shaping and finishing according to the particular product.
Rolling mill	The main machine-based stage; converts pitha/guti into sheet (fali) or circular stock (chak) of required thickness.

6. Bell Metal Production Process



Traditional Sarthebari bell-metal production sequence

The craft combines melting/casting, rolling of stock, and extensive manual forming and finishing.

The exact sequence varies with the product, but the traditional Sarthebari process can be understood through the following main stages:

1. Selection and preparation of raw material: Broken or scrap bell-metal pieces are collected and sorted. Impurities and unsuitable pieces are removed. Where fresh alloying is practised, the copper–tin balance must be controlled because tin content strongly affects hardness and brittleness.

2. Breaking into smaller pieces: Large pieces are reduced to sizes that can be placed conveniently in the melting crucible. Smaller pieces heat more uniformly and melt more readily.

3. Melting in the Muhi: The prepared metal is charged into a crucible called the Muhi and heated in a charcoal-fired furnace. The artisan observes the melt and relies heavily on experience to judge when the material is ready for pouring.

4. Casting in the Aak: The molten metal is poured into a prepared mould called Aak. It solidifies as a thick circular lump or blank locally called Guti or Pitha.

5. Rolling to form Fali or Chak: The Guti/Pitha is taken to a rolling mill and reduced to the required thickness. It may be converted into sheet (Fali) or circular blanks (Chak), depending on the intended article.

6. Heating / softening before forming: The stock is reheated when necessary to make shaping easier. Traditional production depends on repeated control of temperature; overheating or incorrect working temperature can lead to cracking or poor quality.

7. Primary hammer forming: The blank is placed on the anvil (Niyari) and hammered into the basic shape. Several artisans may work together, with the Oja controlling the rhythm, location and force of blows.

8. Intermediate reheating and repeated forming: As the metal work-hardens or cools, it is reheated and hammered again. Repeated cycles gradually produce the required curvature, wall thickness and profile.

9. Fine hammering and dimensional correction: Smaller hammers are used to refine the surface, remove irregularities and bring the vessel or article to a symmetrical final form.

10. Trimming and filing: Rough edges are cut or filed using Khonta and other tools. The rim is made even and safe for handling.

11. Turning / surface finishing: Where required, a hand-operated lathe (Kun) is used for finishing. The surface is scraped, smoothed and polished to develop the characteristic appearance.

12. Inspection and sound/quality check: The artisan checks shape, balance, thickness, defects and, particularly for cymbals, the clarity and persistence of sound. Articles with cracks, poor symmetry or unsatisfactory tone are rejected or reworked.

Why the process needs high skill
High-tin bronze is comparatively hard and can be brittle. The artisan therefore has to control composition, temperature, reheating, hammer force and forming sequence. The craft is not simply “hammering a plate”; it is a carefully managed metallurgical and forming process learned through long practice.

7. Quality, Skill and Craft Characteristics

- Handmade character: slight variations between articles are natural because final shaping is manual.
- Hard, ringing material: the high tin content increases hardness and gives strong reverberation, especially important for tal/cymbals.
- Controlled thickness: uniformity is important for both utensils and musical articles.
- Symmetry and rim quality: uneven rims or distorted geometry reduce both appearance and usability.
- Sound quality: cymbals are judged by tone, loudness, clarity, sustain and absence of cracked or dull sound.
- Surface finish: repeated fine hammering, filing and polishing produce the final visual quality.

8. Advantages of Bell Metal

Advantage	Explanation
High hardness and wear resistance	Tin strengthens copper; high-tin bronze is significantly harder than pure copper and many low-tin bronzes, making utensils and cymbals durable when properly made.
Excellent resonance	The alloy has strong reverberating qualities, making it particularly suitable for bells, cymbals and ritual musical instruments.
Long service life	Well-made articles can remain in household or ceremonial use for generations and are often repaired, reused or recycled.
Attractive appearance	Bell metal has a warm golden/bronze lustre and develops a distinctive traditional surface character.
Cultural value	Objects such as sorai/xorai, bota, kahi and tal have meanings beyond utility and are integral to Assamese ceremonies, hospitality and worship.
Recyclability	Old or broken bell-metal articles can be remelted and returned to the production cycle, which supports a scrap-based traditional economy.
Suitable for thin but strong articles	When correctly processed, the combination of hardness and controlled forming allows relatively thin utensils with good rigidity.
Supports craft-based employment	The production chain sustains Ojas, Bhagias, scrap suppliers,

	charcoal suppliers, rollers, traders and transporters.

9. Disadvantages / Limitations of Bell Metal

Limitation	Explanation
Brittleness at high tin content	Increasing tin raises hardness but reduces ductility; high-tin bronze can crack if worked at unsuitable temperatures or subjected to severe impact.
Difficult manual forming	The alloy is much harder to shape than soft copper. Traditional hammer forming requires repeated heating, skilled coordination and substantial physical effort.
High labour requirement	Most shaping and finishing are manual, so productivity is low compared with mass-produced metal utensils.
Worker drudgery	Repeated hammering, heat exposure, noise and physically demanding postures make the work strenuous.
Dependence on skilled labour	Quality depends heavily on experienced artisans. Loss of hereditary skill or reduced youth participation threatens continuity.
Raw-material cost and variability	A scrap-based system can face price fluctuations and inconsistent composition; alloy variation may affect hardness, cracking and tone.
Higher cost than many factory-made substitutes	Traditional labour, copper/tin cost and low-volume production can make bell-metal products more expensive than steel, aluminium or plastic alternatives.
Surface tarnishing / patina	Like other copper alloys, the surface can darken or develop patina during use and storage; periodic cleaning may be preferred for bright household articles.
Risk of damage if dropped	Hard but brittle articles, particularly thin or highly stressed pieces, may crack rather than bend significantly.

10. Economic and Cultural Importance

- The industry preserves a craft tradition with roots extending back many centuries.
- Bell-metal utensils remain associated with Assamese domestic and ceremonial life.
- Sorai/xorai and bota are symbols of honour, hospitality and religious offering.
- Tal and related cymbals support Vaishnavite devotional music, namghar traditions and folk performance.
- The cluster supports direct artisans and a network of input suppliers and traders.
- Sarthebari products have historically found markets beyond Assam, including other North-Eastern regions and neighbouring Himalayan/Buddhist markets where cymbals are used in worship.

11. Challenges and Future Scope

Although the craft remains culturally important, both official and research literature point to significant production and market challenges. These should be understood separately from the inherent disadvantages of the alloy itself.

Issue / opportunity	Meaning for the craft
Low productivity	Traditional hammer-based methods are slow and physically demanding.

Raw-material prices	Copper/tin and good-quality scrap can be expensive, affecting working capital.
Middlemen and market access	Artisans may receive a smaller share of retail value when dependent on intermediaries.
Youth participation	Younger workers may prefer less strenuous or more secure employment.
Technology adoption	Mechanization must preserve craft quality and employment while reducing drudgery.
Product diversification	Decorative and contemporary utility products can supplement traditional products without replacing them.
Branding and authenticity	Geographical identity, quality assurance, cooperative marketing and digital sales can improve recognition and returns.
Skill documentation and training	Systematic recording of tools, process knowledge and sound/quality criteria can help transmit hereditary knowledge.

12. Important Points

- Bell metal = high-tin bronze, mainly Cu + Sn; commonly about 75–80% Cu and 20–25% Sn in technical descriptions.
- Sarthebari is Assam's best-known bell-metal craft centre and the tradition is traced in official historical accounts to at least the 7th century A.D.
- 1930: Asom Samabai Kohar Sangha helped reduce dependence on Mahajans.
- Traditional workshop = Sal; master = Oja/Kohar; assistants = Bhagias.
- Main products: Kahi, Bati, Ban-bati, Ghoti/Koloh, Bota, Sorai/Xorai, Tema and different Tal/cymbals.
- Muhi = crucible; Aak = mould; Guti/Pitha = thick cast lump; Fali = sheet; Chak = circular blank; Niyari = anvil; Sara = tongs; Khonta = file; Kun = hand-operated lathe.
- Production: scrap preparation → melting → casting → pitha/guti → rolling → heating/forming → fine hammering → filing/finishing → inspection.
- Key advantage: hardness, durability and excellent resonance.
- Key limitation: high-tin bronze is hard but comparatively brittle and difficult to form, so production is labour-intensive and skill-dependent.

Possible Examination Questions

1. Trace the historical development of the bell-metal industry of Sarthebari.
2. Classify the important bell-metal products manufactured at Sarthebari with examples.
3. Explain the organization of a traditional Sal and state the roles of the Oja and Bhagias.
4. Describe the complete production process of a bell-metal utensil in Sarthebari.
5. Write short notes on Muhi, Aak, Pitha/Guti, Fali, Chak, Niyari and Kun.
6. Discuss the advantages and disadvantages of bell metal as an engineering and craft material.
7. Explain the cultural and economic importance of Sarthebari's bell-metal craft.

13. References

1. Commissionerate of Industries & Commerce, Government of Assam. "Present Handicrafts Industries in Assam – Bell Metal Industry of Sarthebari." (Historical background, products, raw materials, technology/tools, employment and marketing).

2. Commissionerate of Industries & Commerce, Government of Assam. “Old Handicrafts Industries in Assam – Brass and Bellmetal Industry.” (Historical patronage, product range and traditional Sal/Oja–Bhagia organization).

3. Sarkar, P. K. and Kakoty, S. K. “Bell Metal Product Manufacturing Techniques: A Technical Report from Sarthebari, Assam.” Journal of The Institution of Engineers (India): Series C, 2021. (Composition, manufacturing and mechanical-property context).

4. Sarkar, P. K. and Kakoty, S. K. “Effect of quenching parameters on mechanical properties of bell metal.” Materials Today: Proceedings, 2021. (High-tin bronze composition and hardness/fracture behaviour).

5. Barpeta District / Government of Assam district industrial material on the Sarthebari bell-metal cluster. (Cluster scale and traditional production characteristics).